



Heaven's light is our Guide

Department of Electrical & Electronic Engineering

Rajshahi University of Engineering & Technology

Design Project on Electrical Service Design (EEE 4100)

Marks		Course Code	EEE 4100
Allocated	Obtained	Course Title	Electrical Service Design
20		Semester & Year	4 th Year Odd Semester (20 series)
		Course Instructor	Prof. Dr. G. K. M. Hasanuzzaman
			Alok Kumar Paul
			Sakib Hossen
		Due Date	
		Submission Date	
Student ID		Student Name	
Submission Deadline		05 November 2025	

General Guideline:

Printed Document:

Print this document and attach your design project at the end of this file before submission.

Viva Voce (If required):

A viva (oral examination) may be conducted to verify or clarify your submitted design project.

Problem statement [CO1-PO(e), CO1-PO(f), CO1-PO(h)]:

Each student must prepare a **complete electrical service design** for a modern apartment building that includes the following components:

1. Conduit Layout

- ✚ Develop detailed conduit routing plans showing the placement, type, and size of conduits.
- ✚ Include power, lighting, and communication circuits.
- ✚ Ensure compliance with BNBC 2020 requirements for installation and accessibility.

2. Fixture Layout

- ✚ Prepare detailed layouts for lighting, ceiling fans, and power outlets.
- ✚ Account for dedicated circuits for heavy-load appliances (e.g., Air conditioner, electric cooking, washer/dryer).

3. Communication Layout

- ✚ Design structured cabling for telephone, internet, intercom, and television systems.
- ✚ Maintain proper separation between communication and power circuits to prevent interference.
- ✚ Indicate communication panel locations and connection points.

4. Lightning Protection System

- ✚ Design a complete lightning protection and grounding network.
- ✚ Include air terminals, down conductors, and earthing arrangements with proper resistance values.
- ✚ Ensure adherence to BNBC and international safety standards.

5. 11kV Substation (Single Line Diagram)

- ✚ Prepare a **Single Line Diagram (SLD)** showing the 11kV substation layout.
- ✚ Include transformer specifications, main switchgear, circuit breakers, and protective devices.

- ✚ Clearly indicate ratings, cable interconnections, and metering arrangements.

6. BOQ/BOM (Bill of Quantities / Bill of Materials)

- ✚ Provide a complete list of materials and components used in the design.
- ✚ Include item descriptions, quantities, specifications, unit prices, and total cost.

Design Requirements

- Select appropriate **cable types, sizes, and protective devices** (MCBs, RCCBs, etc.) based on calculated loads.
- Ensure the total connected load **does not exceed the utility's permitted capacity.**
- Incorporate **energy-efficient and low-power equipment** to promote sustainability.
- Ensure **compliance with safety, environmental, and health standards.**
- Maintain balance between **performance, cost, and sustainability.**

Uniqueness Rules

Last Roll Digit	No. of Floors	Apartments / Floor	Typical Apartment Size	Minimum % of the Air conditioner of the total load
0	8	3	95 m ²	20
1	7	4	90 m ²	25
2	9	3	100 m ²	30
3	10	4	105 m ²	20
4	6	3	85 m ²	25
5	8	4	92 m ²	30
6	9	3	110 m ²	20
7	7	4	88 m ²	25
8	10	3	120 m ²	30
9	6	4	80 m ²	20

** Finally, justify how the corresponding COs and POs are attained through this assignment.

Marking Rubrics:

Criteria (Mapped to POs)	Excellent	Good	Average	Needs Improvement
PO(e): Design, Analysis & Tool Usage (10 Marks)	All layouts (Conduit, Fixture, Communication, Lightning) and the 11kV SLD are technically sound, detailed, accurate, and clearly presented. Load calculations and selection of all components (cables, protective devices) are correct and explicitly justified. (9-10 Marks)	Layouts and SLD are mostly complete and correct but contain minor technical errors, omissions, or lack clarity. Load calculations or component selections have minor errors but are generally appropriate. (7-8 Marks)	Layouts or SLD are incomplete or contain significant technical errors. Load calculations are flawed or key components (e.g., transformer, main switchgear) are incorrectly specified. (5-6 Marks)	Major design components are missing or fundamentally flawed. Layouts are incorrect, and load calculations are absent or fail to meet basic requirements. (3-4 Marks)
PO(f): The Engineer & Society (Safety & Standards) (5 Marks)	The design explicitly and correctly adheres to all required standards, including BNBC 2020, international safety standards, and utility capacity limits. Safety, environmental, and health considerations are clearly integrated. (5 Marks)	The design generally follows standards but has minor deviations or omissions (e.g., minor accessibility issues, incorrect circuit separation). Safety considerations are mentioned but not fully integrated. (4 Marks)	The design overlooks key standards (e.g., BNBC requirements, grounding) or safety factors. Compliance is asserted but not demonstrated in the design. (3 Marks)	The design shows a clear disregard for, or lack of awareness of, fundamental safety standards and legal requirements (BNBC, utility rules). (2 Marks)

PO(h): Ethics, Sustainability & Professionalism (5 Marks)	BOQ/BOM is comprehensive, professional, and accurate, with detailed items, specs, and costs. Sustainability is effectively promoted via energy-efficient equipment, and the balance between cost, performance, and sustainability is well-justified. (5 Marks)	BOQ/BOM is mostly complete but has minor omissions or costing errors. Sustainability is addressed, but the cost/performance balance is not well-justified. (4 Marks)	BOQ/BOM is incomplete, with missing items or specs. Sustainability or energy efficiency is not adequately addressed. (3 Marks)	BOQ/BOM is missing or deeply flawed. There is no consideration for sustainability, cost, or professional ethics. (2 Marks)
Late Submission Penalty	Submitted on time: No Penalty 1 day late: -1 mark > 1 day late: -3 marks			